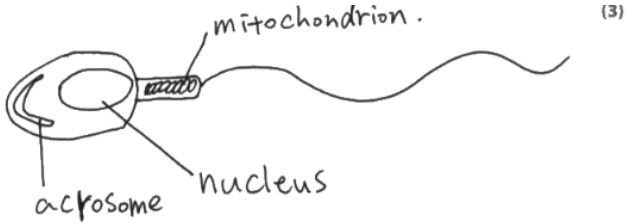


Question Number	Answer	Additional guidance	Mark
4(a)	An answer that includes the following points: • acrosome correctly drawn and labelled (1) • nucleus correctly drawn and labelled (1) • mitochondrion drawn in mid-piece and labelled (1)	correct drawing includes position and suitable shape max 2 for incorrect shape of sperm cell label line must touch structure  mythical diagram (e.g. standard animal cell) scores 1 max for nucleus	(3)

Question Number	Answer	Additional guidance	Mark
4(b)(i)	$= 5.24 \times 10^5 / 524\,000 \text{ (}\mu\text{m}^3\text{)}$	ACCEPT 500 000 to 524 000 5.0×10^5 to 5.24×10^5 max number of 2 dp in standard form	(1)

Question Number	Answer	Additional guidance	Mark
4(b)(ii)	correct ratio (1)	max number of 2 dp 0.06 (: 1) IGNORE 1 : 16.7 ecf applies	(1)

Question Number	Answer	Additional guidance	Mark
4(b)(iii)	An answer that makes reference to three of the following points: • (egg cells are) {haploid / contain one copy of each chromosome} (1) • to ensure the zygote {is diploid / has two copies of each chromosome} (1) • (egg cell) has {cortical granules / zona pellucida} which {prevent polyspermy / additional sperm cells entering egg cell} (1) • (egg cell) has {lipid / oil / fat} (droplets) as a source of energy / mitochondria which {release energy / provide ATP} (1)	only mp2 can be awarded if referring to incorrect gamete ACCEPT so fertilised egg cell {is diploid / has two copies of each chromosome} ACCEPT diploid {zygote / embryo} after fertilisation ACCEPT cortical granules to harden zona pellucida after fertilisation IGNORE jelly coat IGNORE produce energy ACCEPT large so can contain more lipids ACCEPT {lipid (droplets) / protein / carbohydrates} for {production of new cell components / growth of embryo} ACCEPT lipids converted to glucose IGNORE {food / nutrient} stores ACCEPT releases chemicals to attract sperm	(3)

Question Number	Answer	Mark
4(c)	The only correct answer is A 1 and 2 only <i>B is not correct because the cells divide by mitosis</i> <i>C is not correct because the cells divide by mitosis</i> <i>D is not correct because the cells divide by mitosis</i>	(1)